



UNIVERSITÀ DEGLI STUDI DI PISA

CORSO DI STUDIO IN BIONICS ENGINEERING

STATISTICAL SIGNAL PROCESSING - *Prova scritta del 05/11/2024*

PROBLEM

We observe N samples of a realization of the random process $X[n] = A \cdot s[n] + W[n]$, for $n=0, 1, \dots, N-1$, where A is modelled as an unknown deterministic parameter, $s[n]$ is a perfectly known signal with energy $E_s = N$ and $\{W[n]; n = 0, 1, \dots, N-1\}$ are independent and identically distributed (IID) random variables modeling additive white noise, which is independent of A . Each noise sample is Gaussian distributed, with zero mean and variance σ_w^2 . The signal to noise ratio (SNR) is as follows: $\gamma = A^2 / \sigma_w^2$.

- (1) Derive the autocorrelation function (ACF) $R_X[n, m] = E\{X[n]X[n+m]\}$ and establish if $X[n]$ is wide sense stationary (w.s.s.) or not.
- (2) Derive the maximum likelihood (ML) estimator of the unknown parameter A .
- (3) Derive the bias and the mean square error (MSE) of the ML estimator of A , establish if the ML estimator is consistent or not, plot the MSE as a function of γ and find the minimum value of N which guarantees an MSE lower than 10^{-2} when the noise power is $\sigma_w^2 = 2$.
- (4) Derive the Cramér-Rao lower bound (CRB) on the estimate of A and establish if the ML estimator is efficient or not.
- (5) In the case A is modelled as a Gaussian random variable with zero mean and variance σ_A^2 , derive maximum a posteriori (MAP) estimator and express it as a function of the ML estimator, derive the bias and the MSE of the MAP estimator and express it as a function of the MSE of the ML estimator. In this case, the signal to noise ratio (SNR) is as follows: $\gamma = \sigma_A^2 / \sigma_w^2$.



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Solution:

(1) The autocorrelation function (ACF) $R_X[n, m] = E\{X[n]X[n+m]\}$:

$$\begin{aligned} R_X[n, m] &= E\{X[n]X[n+m]\} = E\{(As[n] + W[n])(As[n+m] + W[n+m])\} \\ &= A^2 s[n]s[n+m] + \sigma_w^2 \delta[m] \end{aligned}$$

$X[n]$ is not wide sense stationary (w.s.s.).

(2) The joint pdf of the observed data vector is:

$$X[n] \in N(As[n], \sigma_w^2), \quad \{X[n]\}_{n=0}^{N-1} \text{ indipendenti} \Rightarrow \mathbf{X} \in N(\mathbf{A}\mathbf{s}, \sigma_w^2 \mathbf{I}),$$

where $[\mathbf{s}]_{n+1} = s[n]$, $E_s = \sum_{n=0}^{N-1} s^2[n] = N$.

The pdf in scalar format is:

$$f_X(\mathbf{x}) = \prod_{n=0}^{N-1} f_X(x[n]) = \left(\frac{1}{\sqrt{2\pi\sigma_w^2}} \right)^N \exp\left(-\frac{1}{2\sigma_w^2} \sum_{n=0}^{N-1} (x[n] - As[n])^2 \right)$$

The pdf in vector format is:

$$f_X(\mathbf{x}) = \frac{1}{(2\pi)^{N/2} \sqrt{|\sigma_w^2 \mathbf{I}|}} \exp\left(-\frac{1}{2} (\mathbf{x} - \mathbf{A}\mathbf{s})^T (\sigma_w^2 \mathbf{I})^{-1} (\mathbf{x} - \mathbf{A}\mathbf{s}) \right) = \frac{1}{(2\pi)^{N/2} (\sigma_w^2)^{N/2}} \exp\left(-\frac{1}{2\sigma_w^2} \|\mathbf{x} - \mathbf{A}\mathbf{s}\|^2 \right)$$

Under the assumption that σ_w^2 is a-priori known, the log-likelihood function is:

$$\ln L(A) = \ln f_X(\mathbf{x}) = k - \frac{1}{2\sigma_w^2} \sum_{n=0}^{N-1} (x[n] - As[n])^2 = k - \frac{1}{2\sigma_w^2} \|\mathbf{x} - \mathbf{A}\mathbf{s}\|^2$$

The ML estimate of A is given by:

$$\frac{d \ln L(A)}{dA} = \frac{1}{\sigma_w^2} \sum_{n=0}^{N-1} s[n] (x[n] - As[n]) = 0 \Rightarrow \hat{A}_{ML} = \frac{\sum_{n=0}^{N-1} s[n]x[n]}{\sum_{n=0}^{N-1} s^2[n]} = \frac{1}{N} \sum_{n=0}^{N-1} s[n]x[n] = \frac{1}{N} \mathbf{s}^T \mathbf{x}$$

Since the estimate is a linear function of the data (in practice, it is the output of the linear time-invariant matched filter), which are Gaussian distributed, the estimate is also Gaussian distributed.



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(3) Bias and the mean square error (MSE) of the ML estimator of A .

$$b\{\hat{A}_{ML}\} = E\{A - \hat{A}_{ML}\} = A - E\{\hat{A}_{ML}\} = A - \frac{A}{N} \mathbf{s}^T \mathbf{s} = 0$$

$$MSE\{\hat{A}_{ML}\} = \text{var}\{\hat{A}_{ML}\} = E\left\{\left(\frac{1}{N} \mathbf{s}^T \mathbf{w}\right)^2\right\} = \frac{1}{N^2} E\{\mathbf{s}^T \mathbf{w} \mathbf{w}^T \mathbf{s}\} = \frac{\sigma_w^2}{N^2} \mathbf{s}^T \mathbf{s} = \frac{\sigma_w^2}{N}$$

The estimator is unbiased and consistent.

$$MSE\{\hat{A}_{ML}\} = \frac{\sigma_w^2}{N} < 10^{-2} \rightarrow N > 10^2 \sigma_w^2 = 200$$

(4) The Cramér-Rao lower bound (CRB).

$$\frac{d^2 \ln L(A)}{dA^2} = \frac{d}{dA} \left(\frac{1}{\sigma_w^2} \sum_{n=0}^{N-1} s[n](x[n] - As[n]) \right) = -\frac{1}{\sigma_w^2} \sum_{n=0}^{N-1} s^2[n] = -\frac{N}{\sigma_w^2}$$

$$CRB(A) = \frac{\sigma_w^2}{N} = MSE\{\hat{A}_{ML}\}$$

The estimator is efficient.

(5) MAP estimator of A .

The joint pdf of $\mathbf{x}$ and A in scalar format is:

$$\begin{aligned} f_{X,A}(\mathbf{x}, A) &= f_{X|A}(\mathbf{x}|A) f_A(A) = \prod_{n=0}^{N-1} f_{X|A}(x[n]|A) f_A(A) \\ &= \left(\frac{1}{\sqrt{2\pi\sigma_w^2}} \right)^N \exp\left(-\frac{1}{2\sigma_w^2} \sum_{n=0}^{N-1} (x[n] - As[n])^2 \right) \frac{1}{\sqrt{2\pi\sigma_A^2}} \exp\left(-\frac{A^2}{2\sigma_A^2} \right) \end{aligned}$$

$$\ln f_{X,A}(\mathbf{x}, A) = \ln f_{X|A}(\mathbf{x}|A) + \ln f_A(A) = k - \frac{1}{2\sigma_w^2} \sum_{n=0}^{N-1} (x[n] - As[n])^2 - \frac{A^2}{2\sigma_A^2}$$

The MAP estimate of A is obtained as follows:

$$\begin{aligned} \frac{d \ln f_{X,A}(\mathbf{x}, A)}{dA} &= \frac{1}{\sigma_w^2} \sum_{n=0}^{N-1} s[n](x[n] - As[n]) - \frac{A}{\sigma_A^2} \\ &= \left(-\frac{1}{\sigma_A^2} - \frac{N}{\sigma_w^2} \right) A + \frac{1}{\sigma_w^2} \mathbf{s}^T \mathbf{x} = -\left(\frac{\sigma_A^2 N + \sigma_w^2}{\sigma_A^2 \sigma_w^2} \right) A + \frac{1}{\sigma_w^2} \mathbf{s}^T \mathbf{x} = 0 \end{aligned}$$



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$$\hat{A}_{MAP} = \frac{1}{\sigma_W^2} \mathbf{s}^T \mathbf{x} \left(\frac{\sigma_A^2 \sigma_W^2}{\sigma_A^2 N + \sigma_W^2} \right) = \frac{1}{N} \mathbf{s}^T \mathbf{x} \left(\frac{N\gamma}{N\gamma + 1} \right) = \alpha \hat{A}_{ML}$$

$$\text{where } \gamma = \frac{\sigma_A^2}{\sigma_W^2}, \quad \alpha = \frac{N\gamma}{N\gamma + 1}$$

$$b\{\hat{A}_{MAP}\} = E\{A - \hat{A}_{MAP}\} = E\{A\} - E\{\hat{A}_{MAP}\} = E\{A\} - \alpha E\{A\} = 0$$

$$\begin{aligned} MSE\{\hat{A}_{MAP}\} &= E\left\{\left(A - \hat{A}_{MAP}\right)^2\right\} = E\left\{\left(A - \alpha \hat{A}_{ML}\right)^2\right\} \\ &= E\left\{\left(A - \alpha A + \alpha A - \alpha \hat{A}_{ML}\right)^2\right\} = E\left\{\left[\alpha(A - \hat{A}_{ML}) + (1 - \alpha)A\right]^2\right\} \\ &= \alpha^2 E\left\{\left(A - \hat{A}_{ML}\right)^2\right\} + (1 - \alpha)^2 E\{A^2\} + 2\alpha(1 - \alpha)E\left\{\left(A - \hat{A}_{ML}\right)A\right\} \\ &= \alpha^2 \frac{\sigma_W^2}{N} + (1 - \alpha)^2 \sigma_A^2 + 2\alpha(1 - \alpha)E\left\{\frac{2}{N} \mathbf{c}^T \mathbf{W} A\right\} = \alpha^2 \frac{2\sigma_W^2}{N} + (1 - \alpha)^2 \sigma_A^2 \\ &= \frac{\sigma_W^2}{N} \left(\alpha^2 + (1 - \alpha)^2 N\gamma\right) = \frac{\sigma_W^2}{N} \left(\frac{(N\gamma)^2}{(N\gamma + 1)^2} + \frac{N\gamma}{(N\gamma + 1)^2}\right) \\ &= \frac{\sigma_W^2}{N} \left(\frac{N\gamma}{N\gamma + 1}\right) = \alpha \frac{\sigma_W^2}{N} \end{aligned}$$

$$MSE\{\hat{A}_{MAP}\} = \alpha \frac{\sigma_W^2}{N} = \alpha MSE\{\hat{A}_{ML}\} \leq MSE\{\hat{A}_{ML}\}$$